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Effectively and Efficiently Supporting Predictive Big Data Analytics over Open Big Data in the Transportation Sector: A Bayesian Network Framework

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***Abstract*—Today, various types of valuable data can be collected with ease and at a rapid pace. In recent years, many governments, researchers, and organizations have been driven by open data pioneers, to make their data available for public. Transportation data, such as public bus performance data, is an example of open big data. The analyzing of these open big data can be used in social services. For example, bus service operators might get a vision into time delays in bus services by processing and mining public bus performance data. Then, making ameliorative steps (e.g., adding more buses, rerouting some bus routes, etc.) results in improving the feeling of the passenger. We provide a *Bayesian framework*, which is applied on big data obtained from transportation system. Specifically, a number of *Bayesian networks* have been used in our framework to predict whether a bus will arrive late or early at a specific bus stop. We investigate and establish the optimum network settings and/or parameter permutations for each (bus stop, bus route, arrival time)-triplet. The results demonstrate that the proposed *Bayesian framework* effectively supports predictive analytics on big transportation data collected from the City of Winnipeg, Manitoba, Canada.**

Keywords—big data, open big data, big data management, big data analytics, big data applications, big data systems

I. Introduction

Today, various types of valuable big data [1, 2] can be collected with ease and at a rapid pace. In recent years, many governments, researchers, and organizations have been driven by open data pioneers, to make their data available for public. Some instances of open big data are:

- biodiversity data [3, 4],
- census data [5],
- financial time series [6-10],
- graphs [11],
- healthcare and disease reports (e.g., coronavirus disease 2019 (COVID-19) data) [12-21],
- patent register [22],
- social networks [23-27],
- trajectories [28],
- transportation data (e.g., public bus performance data) [29-31], and
- weather data [32, 33].

The use of open big data can be beneficial for social purposes. For example, analyzing and processing of the public bus performance data helps to obtain a beneficial vision on transportation delays that can lead the transportation service managers to improve the performance. As a result, the passenger pleasure might be improved by adopting suitable steps (e.g., adding more buses, rerouting some buses routes, etc.).

In present cities, public transit plays a critical role. There are many advantages for performance optimization of public transportation services including:

- decreasing traffic and pollution,
- financial return for the city,
- etc.

Reduced transportation costs and greater safety would benefit public transit users in an optimized public transport network.

Scheduling times are one of the most important areas where productivity is lost. Although, the precise schedule deviation for each city is different from another, but the instances of them show the efficiency. For instance, one study[1] reported that on average 8% increase occurred in passengers' travel time, because of the arrival delays in the New York City subway system. This higher cost has consequences; it reduces public trust in the transportation system, with unreliability noted as a

[1] https://ny.curbed.com/2017/10/13/16455880/ new-york-subway-mta-operating-cost-analysis

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key reason in potential passengers preferring for alternate types of transit[2]. Therefore, a reliable method is necessary to identify areas of a public transit system where chronic delays are occurring so that improvement efforts can be focused on them.

To address it, a *Bayesian framework* is proposed in this paper to support the predictive analytics using big data acquired from transportation system. This framework uses a number of *Bayesian networks* to predict if a bus will arrive at a bus stop later than expected. Analysis and parameter permutation are applied to achieve the best result for each

(bus stop, bus route, arrival time)-triplet.

The efficiency of our *Bayesian framework* in predictive analytics on open data for transportation analytics via machine learning is demonstrated from analysis on an open big transportation data collected from the Canadian city of Winnipeg.

Our Bayesian approach for conducting predictive analytics over massive transit data is one of the paper's main targets. An ensemble of *Bayesian networks* is used to predict on-time performance or lateness of buses at a given bus stop at a predetermined time. The practicality of the framework is shown using results obtained from experiments on real-life bus transportation data.

The rest of this paper is organized in the following manner. The next section presents background and related works. Section III describes our framework. Sections IV and V show evaluation and conclusions, respectively.

II. BACKGROUND

A. Related Work

There are big data examples (e.g., big transportation data) everywhere. Data sciences [33, 34], analytics [35-37], and technologies make appropriate use of:

- data mining methods [38-41],
- machine learning tools [42-45], and/or
- mathematical and statistical modeling [46].

They can all be used to identify relevant information and useful knowledge existed in the big data. The popularity of open data has increased within the past few years because it allows governments to provide citizens with transparency about government activities, and it allows citizens to obtain some insight into the government-supported services. As a preview, open transit data (especially, public transportation data) is investigated in this study.

It has become demanding to process big transit data. Several researchers, e.g. Zhao *et al.* [47], proposed a model to predict the demand for private vehicles (e.g., taxis) and ride-sharing services (e.g., *Uber*). To produce reliable predictions, the temporal-correlated entropy is used to assess demand and is included into many predictor algorithms such as the *Markov chain stochastic model* and the *long short-term memory* (LSTM) algorithm. Zhao *et al.* conducted research on predicting usage rate for private transportation services, on the contrary, we analyze the usage for public vehicle services, particularly bus services.

There have been several studies that analyze public transportation data in order to predict the arrival time of buses or delays. For instance:

- Audu *et al.* [29] precisely estimated bus arrival time delays using available data from the Canadian city of Toronto.
- Yamaguchi *et al*. [48] exploit a month of bus investigate data from Fukuoka, Japan, to predict bus lag time using several complex machine learning models such as *artificial neural networks* (ANN) and *gradient boosting decision trees* (GBDT).
- To predict bus arrival times in the Chinese city of Beijing, Liu *et al.* [49] have adapted a modified *K-Nearest Neighbor* method.

In contrast, a machine learning model of *Bayesian networks* is adapted for our predictive approach; then, our approach is examined by open data from another Canadian city—Winnipeg in the Canadian Prairies in Central Canada.

There is a strong bias on passenger data in the available research on public transit. Data on passenger numbers, tracking, and other use variables have been available for previous research with the operating authorities [50-52]. We intend to lead our work to the approach totally related to the bus transportation data. As a result, our analysis is protected from any privacy worries. Additionally, due to the *Google general transit feed specification* (GTFS) (https://developers.google.com/transit/gtfs), making any changes can be done in our research methodology. Additionally, *Bayesian networks* are not widely used for bus data, and this study attempts to fill this gap.

Open transportation data is examined to investigate area of the public bus-transportation system concerning chronic lateness or deviation from planned schedules. This will be achieved by establishing a *Bayesian network* from the acquired data, that will be used for prediction of the bus delay.

B. Fundation

Since a *Bayesian network* is a structured graph representation of probabilistic connections between numerous random variables, a missing line between two random variables reflects conditional independence. If $P(X \cap Y) = P(X)P(Y)$, then the two random variables X and Y are defined as independent variables, according to statistics or probability theory.

Moreover, if the following two conditions are met, three random variables *X, Y*, and *Z* are shown to be independent:

$$\begin{gathered} P(X \cap Y) = P(X)P(Y) \\ P(X \cap Z) = P(X)P(Z) \\ P(Y \cap Z) = P(Y)P(Z) \end{gathered} \tag{1}$$

[2] https://www.forbes.com/sites/jeffmcmahon/2013/03/06/ top-eight-reasons-people-give-up-on-public-transit/

$$P(X \cap Y \cap Z) = P(X)\, P(Y)\, P(Z) \tag{2}$$

The constraint for the conditional independence of the two random variables X and Y relative to the third variable Z is as follows:

$$P(X \cap Y|Z) = P(X|Z)\, P(Y|Z) \tag{3}$$

or equivalently,

$$P(X|Y \cap Z) = P(X|Z) \tag{4}$$

In the *Bayesian network* or graph representation, the random variables are represented as nodes and the direct dependencies between variables are represented as edges. If there is an edge (X, Y) in the graph linking two random variables X and Y, then $P(X \mid Y)$ is a factor in the joint probability distribution [53]. As such, all values of X and Y require $P(Y \mid X)$ to make inference.

A local Markov property is satisfied in *Bayesian networks* [53], which states that given a node's parents, it is conditionally independent of its non-descendants. Using the *Chain Rule of Probability*, the property may be used to simplify the joint distribution.

In more detail, the Chain Rule of Probability can be used to give the joint probability distribution of random variables $X_1, X_2, \ldots, X_n$ [54, 55]:

$$P(\cap_{k=1}^{n} X_k) = \prod_{k=1}^{n} P(X_k \mid \cap_{j=1}^{k-1} X_j) \tag{5}$$

After reduction, the joint distribution for a *Bayesian network* is resulted as follows:

$$P(A_1, A_2, \ldots, A_n) = \prod_{i=1}^{n} P(A_i | Parents(A_i)) \tag{6}$$

Since most nodes will have few parents in comparison to the overall size of the network, the property can be applied in large networks to substantially decrease the computation cost.

III. Bayesian Network Framework for Supporting Predictive Big Data Analytics over Open Big Data in the Transportation Sector Effectively and Efficiently

To construct our *Bayesian framework*, we go through a set of steps aimed to provide a clear overview of the goal sought and the methods utilized to achieve it:

1. *Data acquisition* is considered as the first stage. This is the process of developing a suitable data set. A robust and easily expandable dataset should be selected. A robust dataset has all of the information we need to evaluate without any need for additional datasets.
2. *Feature selection* is considered as the next stage. Features such as bus arrival times, scheduled bus stops, bus stop locations, journey times, Global Positioning System (GPS) coordinates, and so on are included. We can update the model as new data is received, allowing us to further refine the model rather than rebuilding as each new piece of information comes in.
3. *Data preprocessing* is considered as the next stage. In this stage, some steps including data filtering or format modification are applied on the acquired datasets in order to extract useful information easier.
4. *Data analysis* is the primary focus of our project. Non-obvious correlations and patterns within the preprocessed data is revealed by adapting a series of data mining and machine learning techniques—e.g., *Bayesian networks*—using the preprocessed data.
5. *Evaluation* is the last stage. Our work is assessed by a number of predictions we make, which we are able to compare it with future outcomes.

A. Data Collection

For data collection, we acquire data from open data portals from many cities in the world. Examples are:

- Montreal, Toronto, Vancouver, and Winnipeg, all in Canada;
- New York, NY, in the USA;
- etc.

Several features are included in dataset. Examples are:

- information pertaining to agency information and contact,
- operation calendar (e.g., weekly, seasonal),
- fare pricing,
- scheduled routes (including sequence of physical stops, and start/end destinations), and
- bus stop location data.

These basic datasets are formatted according to the GTFS standard.

An *application programming interface* (API) enables users to obtain real-time data from open data portals. For instance, a file containing all salient information regarding a bus stop in real-time, may be returned from the API, by making an appropriate call, as can be seen in Fig. 1. The file can be in different formats:

- comma-separated values (CSV),
- eXtensible Markup Language (XML), and
- JavaScript Object Notation (JSON).

Fig. 1 illustrates that, the data includes a lot of important information, such as bus stop schedules. The data for a bus stop includes:

- A bus stop identifier, such as:
  - name (e.g., westbound Grant at Oakdale),
  - bus stop number (e.g., 60535), and
  - its Global Positioning System (GPS) location in the form of geographic latitude and longitude (e.g., (49.85946, –97.26983)).

- A list of the information of the buses which are serving the bus stop, such as:
  - bus identifying number (e.g., 66),
  - route name (e.g., Route 66 Grant), and
  - a list of scheduled stops. Data includes scheduled arrival time and the estimated arrival time, for each scheduled stop. For example, in the case of arrival time it defined as 2018-11-14 T14:42:42, which refers to 14h 42m 42s (i.e., 2:42:42 pm) on November 14, 2018, and in the case of the estimated arrival time it defined as 2018-11-14 T14:48:13, which refers to 14h 48m 13s (i.e., 2:48:13 pm) on the same day.


```
"stop-schedule": {
    "stop": {
        "key": 60535,
        "name": "Westbound Grant at Oakdale",
        "number":     60535,
      "direction":   "Westbound",
      "side":        "Nearside",
      "street": {...},
        "cross-street":
      {...},"centre": {
          "utm": {...},
          "geographic": {
            "latitude": "49.85946",
            "longitude": "–97.26983"}}},
        "route-
    schedules": [
      {"route": {
          "key": 66,
          "number": 66,
          "name":   "Route   66
       Grant",    "customer-type":
       "regular",       "coverage":
       "regular"},
        "scheduled-stops": [
          {"key": "9839611-51",
            "cancelled":
        "false","times": {
              "arrival": {
                 "scheduled":"2018-11-14
              T14:42:42",   "estimated":"2018-11-14
              T14:48:13"},
              "departure": {...}},
            "variant": {...},
            "bus": {...}},
          ...]},
        ...,
        {"route": {
          "key": 65,
          "number": 65,
          "name":   "Route   65   Grant
       Express",        "customer-type":
       "regular", "coverage": "express"},
        "scheduled-stops": [
          {"key": "9839526-24",
            "cancelled":
        "false","times": {
              "arrival": {
                 "scheduled": "2018-11-14 T15:47:31",
                 "estimated": "2018-11-14 T15:47:31"},
              "departure": {...}},
            "variant": {...},
            "bus": {...}},
          ...]},
        ...]},
    "query-time": "2018-11-14T14:44:09"
```


Fig. 1. A sample of bus data.

Each API call returns both the (scheduled and estimated) arrival time of the next bus and the few next buses, serving just one stop. However, making multiple API calls is needed to request and collect data for all bus stops. Besides, to optimize the bandwidth needed for their service, some data portals need all API calls to be represented with a key. The maximum number of requests per minute each key can make is limited. For instance, we need 52 keys to request and collect data for near 5,187 bus stops served by Winnipeg transportation, due to its data portal allows only 100 API requests per minute per key.

B. *Initial Data Manipulation*

We preprocess the data in two stages following data collection:

- The first phase entails *fetching essential information from the acquired data*, such as:
  - bus stop identifier,
  - GPS location (in the form of geographic latitude and longitude),
  - route name,
  - scheduled arrival, and
  - arrival difference.

  Here, the lateness of bus is captured by the arrival time, which is computed by the difference between the estimated and scheduled arrival times:

$$\begin{aligned} &\textit{arrival difference} \\ &= \textit{estimated arrival time} \\ &\quad - \textit{scheduled arrival time} \end{aligned} \tag{7}$$

  According to this equation, if a bus arrival delays then we have a positive arrival difference. Buses are more likely to be late with a larger arrival difference. On-time arrival for a bus is considered as a zero arrival difference. For the sake of practicality, if a bus arrival encountered a small delay (e.g., within 3 min of its scheduled arrival time) it is usually spotted as on time.

  In order to be clear, a negative difference in arrival time represents a bus arrives early. Again, from a practical viewpoint, the state of on-time bus arrival indicates that a bus arrives slightly early (e.g., within 1 min of its scheduled arrival time).

- The *collection of essential statistical data from our dataset* is the second critical stage in preprocessing. We accomplish this by performing two basic functions:
  - First, on-time arrival is filtered out and only early or late arrival is considered.
  - Then, by utilizing a map data structure, we calculate the overall observed frequency of each data point parameter.

  In this way, a data element can be quickly tested for membership and its frequency can be looked up. The relative-error based evaluation methods is used in the next analysis. Both procedures are running together.

C. Data Analytics

We analyze data after they have been preprocessed. Specifically, in our work, the three variables—namely, the route lateness, bus stop lateness and arrival time deviation—are considered. Then, we produce the conditional probabilities and joint distribution of the aforementioned variables using the following three parameters:

- δ: "There is a late bus at bus stop δ";
- λ: "A scheduled stop on bus route λ is late";
- μ: "A bus scheduled to arrive at time μ is late".

For example:

- $P(\delta = 10093)$ is the probability of being late at bus stop 10093;
- $P(\lambda = \text{'Route 51'})$ corresponds to the probability of being late on route 51;
- $P(\mu = 2{:}00\ \text{pm})$ corresponds to the probability of being late at 2:00 pm.

A given triplet (δ, λ, μ) is considered to find the conditional dependencies between the random variables *X, Y* and *Z*. The probabilities of the joint distribution can be calculated as follows:

$$\begin{gathered} P(X) \\ P(X \cap Y) \\ P(X \cap Y \cap Z) \\ P(X|Y) \\ P(X|Y \cap Z) \\ P(X \cap Y|Z) \end{gathered} \tag{8}$$

where *X, Y* and *Z* have different types of variables. The independence between the random variables *X, Y* and *Z* are examined. The five possible *Bayesian network* configurations is produced using these three variables, as can be seen in Fig. 2:

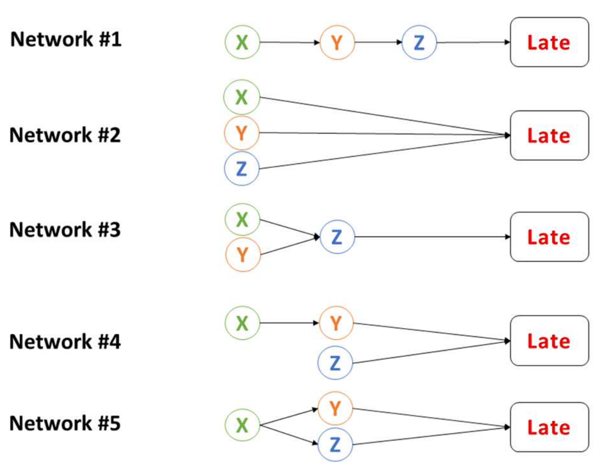

Fig. 2. Bayesian network configuration.

- Network #1, which is a basic serial network. The random variable *Y* is defined by *X*, which in turn *Z* is defined by *Y*.
- Network #2, where the random variables *X, Y*, and *Z* are established conditionally independent from each other. This can be considered as three parallel serial networks or three independent probabilities.
- Network #3, in which the both random variables *X* and *Y* are dependent from *Z*.
- Network #4, where includes two parallel serial networks:
  - (*X*→*Y*), and
  - *Z*.
- Network #5, which is a diamond network. The random variable *X* is considered as parent for *Y* and *Z*, but both *Y* and *Z* are conditionally independent of each other.

IV. Experimental Assessment and Analysis

After the generating of our *Bayesian framework* which includes two networks for data analysis, then the evaluation of these networks is performed. For each

(bus stop δ, bus route λ, arrival time μ)-triplet,

a *Bayesian network* is constructed. Then, we compare the resulting probability with the one pre-computed from the data. All the five networks settings were evaluated by each data triplets, in all 3! = 6 permutations. Then, we stored for each triplet, the network and the parameter permutation that gained the best performance.

The real-life public transportation data from Winnipeg Transit Open Data portal (https://winnipegtransit.com/en/open-data/) is used to build and evaluate our *Bayesian framework*. Thus, the data acquisition performed in these steps:

- downloading the static Winnipeg Transit GTFS dataset (http://gtfs.winnipegtransit.com/google_transit.zip); and
- generating several calls to its Stop Schedule API (among the Open Data Web Service API (https://api.winnipegtransit.com/home/api/v3)) to acquire the real-time schedule information for the 5,187 bus stops served by Winnipeg Transit.

Then, after preprocessing of the real-life data, we defined whether a bus arrives early or late. Afterwards, the data analysis was conducted by training *Bayesian framework* with the early and late bus arrival data, and testing the framework with unseen data.

Fig. 3 demonstrates the performance results of our framework in predictions. The figure illustrates a good performance of our framework. A 'best' network 93.71% of the time was generated by each triplet, with a relative error target of 1%.

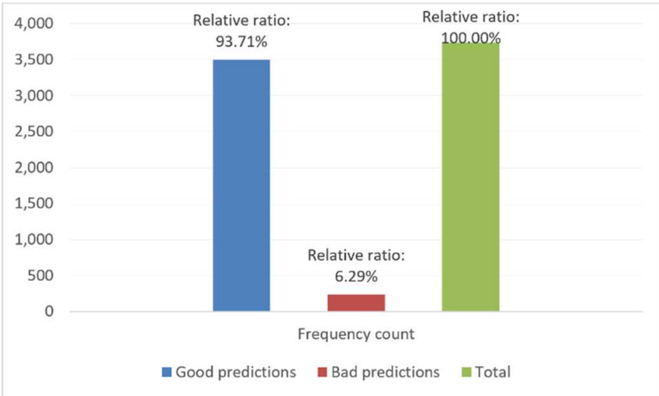

Fig. 3. A summarization of the full output.

Fig. 4 shows the results of the five network configurations (X=bus stop, Y=bus route, Z=arrival time). Among them:

- network #3: {X & Y → Z}, and
- network #1: {X → Y → Z}

resulted as two 'best' network configurations, with them accounting for 53.37% and 38.06% of the 'best' networks respectively. Consider that, in the network #3, there exists a dependency of Z to both X and Y (i.e., arrival time is dependent on both bus stop and route). Network #1 is a serial network.

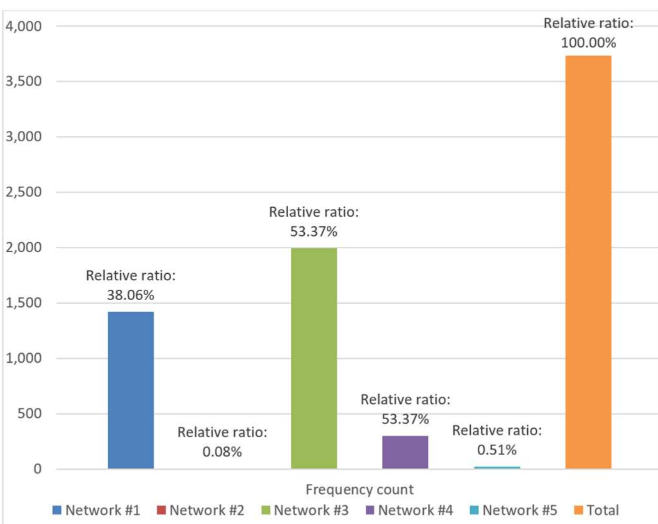

Fig. 4. Count for each network configuration.

The results of six network permutations were shown in Fig. 5, similar to our results on the 'best' network among the five configurations. According to the results, the two permutations were remarkable:

- *ZXY* (arrival-stop-route), and
- *ZYX* (arrival-route-stop),

which resulted 48.55% and 43.04% of 'best' permutations respectively.

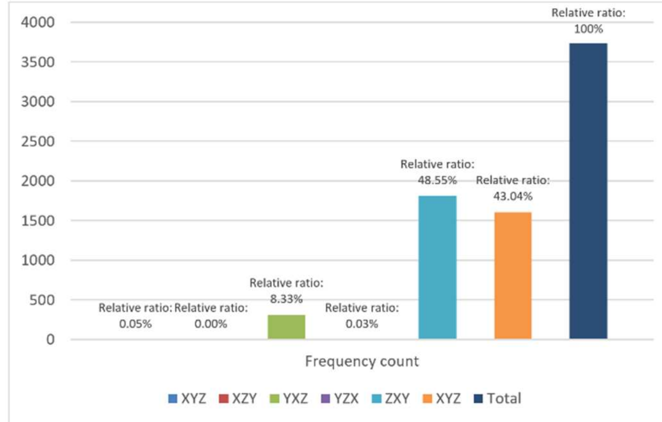

Fig. 5. Count for each operand order.

V. Conclusions and Future Work

Today, various types of valuable data can be collected with ease and at a rapid pace. In recent years, many governments, researchers, and organizations have been driven by open data pioneers, to make their data available for public. Public bus performance data as an example of transportation data, for instance, is an open big data. The analyzing of these open big data can be used in social services. For instance, an insight on the on-time performance or delay in bus services gives is obtained the bus service providers, by analyzing and mining the public bus performance data. Thus, the information helps to be more prepared by doing actions such as adding more buses, rerouting some buses routes, etc. So, the passengers feel more reliability.

In this paper, we propose a *Bayesian framework* used to improve predictive analytics over big transportation data from the City of Winnipeg, Manitoba, Canada. Specifically, five network configurations were established in our framework to predict whether a bus would arrive late at a given bus stop for a scheduled time. We obtained six parameter permutations, by considering the three parameters including bus stop, bus route and bus arrival time. We analyzed and determined the network configurations and/or parameter permutation to produce the best result for each

(bus stop, bus route, arrival time)-triplet.

The effectiveness and practicality of our *Bayesian framework* in supporting predictive analytics over big open data for transportation analytics via machine learning was demonstrated by evaluation on a real-life open big data from a Canadian city of Winnipeg shows. Particularly, the results show that the most important parameter in predicting bus delay is the arrival time.

Understanding the many variables that contribute to bus lateness may be a very useful tool. As a result of our technique, bus scheduling was improved and the ideal number of buses that should be transiting concurrently was discovered, significantly reducing bus lateness. Generating a *Bayesian network* to define the conditional dependencies between the variables such as route lateness, bus stop lateness and expected arrival time deviation can be exploited among the different values of each random variable, to compute the correlations relationships. Furthermore, while we showed our approach using bus data in a

Canadian city, the knowledge gained and the solution are likely to be adaptable to other modes of transportation and/or other cities through transfer learning.

Further investigating correlations between the lateness of several different bus stops (e.g., Are two bus stops located in the same neighborhood more likely to have similar lateness?) and/or routes (e.g., How to predict the lateness on route *A* knowing the lateness on route *B*?) can be done as ongoing and future research.

Besides, to achieve a deeper understanding of the logic of bus lateness, we investigate the modulation of other open big data (e.g., weather, traffic condition, passenger count) into our framework. We also intend to improve our framework by allowing more dynamically formed networks, on-the-fly insertion of additional variables, and/or recalculations of probabilities using streaming input data. Incorporating relevant techniques (e.g., data cube and OLAP [56-60]) into our framework in supporting predictive analytics over big transportation data, also can be performed.

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